APP.PY

```
import os

import json

import binascii

from flask import Flask, render_template, request, flash, redirect, url_for,
send_from_directory

from werkzeug.utils import secure_filename

from utils.crypto import AESCipher

from utils.binary import bytes_to_binary, binary_to_bytes

from utils.steganography import embed_secret, extract_secret


app = Flask(__name__)

app.secret_key = 'super_secret_key_for_session'


# Configuration

UPLOAD_FOLDER = 'uploads'

ALLOWED_IMAGE_EXTENSIONS = {'png'}

ALLOWED_DOC_EXTENSIONS = {'pdf', 'txt'}

MAX_FILE_SIZE = 1 * 1024 * 1024  # 1 MB


app.config['UPLOAD_FOLDER'] = UPLOAD_FOLDER


def allowed_file(filename, allowed_extensions):

    return '.' in filename and \

        filename.rsplit('.', 1)[1].lower() in allowed_extensions


@app.route('/')
```

```
def home():
    return render_template('index.html')

@app.route('/upload', methods=['POST'])
def upload_files():
    if 'cover_image' not in request.files or 'secret_file' not in request.files:
        flash('No file part', 'error')
        return redirect(request.url)

    cover_image = request.files['cover_image']
    secret_file = request.files['secret_file']

    # 1. Validation
    if cover_image.filename == '' or secret_file.filename == '':
        flash('Files not selected', 'error')
        return redirect(request.url)

    if not allowed_file(cover_image.filename, ALLOWED_IMAGE_EXTENSIONS):
        flash('Invalid image format. Only PNG is allowed.', 'error')
        return redirect(request.url)

    if not allowed_file(secret_file.filename, ALLOWED_DOC_EXTENSIONS):
        flash('Invalid secret file format. Only PDF or TXT allowed.', 'error')
        return redirect(request.url)

    # 4. Save and Start Processing
    image_filename = secure_filename(cover_image.filename)
    secret_filename = secure_filename(secret_file.filename)
```

```
cover_path = os.path.join(app.config['UPLOAD_FOLDER'], image_filename)
secret_path = os.path.join(app.config['UPLOAD_FOLDER'], secret_filename)

cover_image.save(cover_path)
secret_file.save(secret_path)

# 5. Encryption & Embedding
aes = AESCipher()
encrypted_bytes, iv, key = aes.encrypt_file(secret_path)
raw_binary_string = bytes_to_binary(encrypted_bytes)

# Error Correction
from utils.binary import add_error_correction
encoded_binary_string = add_error_correction(raw_binary_string, n=5)

# Save Metadata
metadata = {
    'original_filename': secret_filename,
    'key': binascii.hexlify(key).decode('utf-8'),
    'iv': binascii.hexlify(iv).decode('utf-8'),
    'binary_length': len(encoded_binary_string)
}

metadata_path = os.path.join('processed', f"{secret_filename}_meta.json")
with open(metadata_path, 'w') as f:
    json.dump(metadata, f, indent=4)
```

```
    # Generate Stego Image
    stego_filename = f"stego_{image_filename}"
    stego_path = os.path.join('processed', stego_filename)
    embed_secret(cover_path, encoded_binary_string, stego_path)

    metadata['stego_image_path'] = stego_path
    with open(metadata_path, 'w') as f:
        json.dump(metadata, f, indent=4)

    flash(f'Steganography Complete!', 'success')
    return render_template('index.html', stego_filename=stego_filename)

# Download Route
@app.route('/download/<filename>')
def download_file(filename):
    return send_from_directory('processed', filename, as_attachment=True)

if __name__ == '__main__':
    app.run(debug=True, use_reloader=False)
```

UPLOAD PAGE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Secure Steganography</title>
  <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>
<body>
  <div class="container">
    <h1>Secure Steganography</h1>

    <div class="nav-tabs">
      <a href="/" class="active">Hide Data</a>
      <a href="/extract-page">Extract Data</a>
    </div>

    {% with messages = get_flashed_messages(with_categories=true) %}
     {% if messages %}
      {% for category, message in messages %}
        <div class="alert {{ category }}">{{ message }}</div>
      {% endfor %}
     {% endif %}
    {% endwith %}
```

```
    {% if stego_filename %}
    <div class="result-card">
      <h3>Process Complete!</h3>
      <p>Your data has been securely hidden.</p>
      <a href="{{ url_for('download_file', filename=stego_filename) }}" class="btn" style="background:#27ae60;">Download Stego Image</a>
    </div>
    {% else %}

    <form action="/upload" method="post" enctype="multipart/form-data" class="upload-form">
      <div class="form-group">
        <label for="cover_image">1. Select Cover Image (PNG):</label>
        <input type="file" name="cover_image" id="cover_image" accept=".png" required>
      </div>

      <div class="form-group">
        <label for="secret_file">2. Select Secret File (PDF/TXT, Max 1MB):</label>
        <input type="file" name="secret_file" id="secret_file" accept=".pdf, .txt" required>
      </div>

      <button type="submit" class="btn">🚀 Hide Data Securely</button>
    </form>
    {% endif %}
  </div>
</body>
```

</html>